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India Smart Grid Forum

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JVVNL

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DISTRIBUTION UTILITY MEET DUM 2026

KNOWLEDGE PARTNER

Power Foundation of India

CO HOST UTILITIES

TATA POWER

TATA POWER-DDL Towards a Greener Tomorrow

Chandigarh Power Distribution Limited (CPDL)

BSES Rajdhani Power Limited

BSES Yamuna Power Limited

NPCL Noida Power Company Limited

DUM 2026 CONFERENCE SESSION

Session 6: India Energy Stack (IES), Data Exchange Framework, Cyber Security and Data Privacy for the Digital Grids

India Energy Stack: Secure Data Exchange for the Digital Grid

Presented By

Prince Dhawan, IAS, Executive Director, REC Limited

Distribution Utility Meet | 27 - 28 October 2026 | www.dumindia.in



CONTEXT

The gap is coordination, not technology

22.44 Cr
smart meters sanctioned

500 GW
non-fossil capacity by 2030

100+
utilities across States and UTs

DISCOMs cannot see rooftop solar

Its data goes to OEM clouds

Consumers cannot share their own data

No consent-based way to do it

Every new partner needs a new integration

And it is rebuilt at every refresh

RELEVANCE

Why this matters for DISCOMs now

Integration cost

Build once, not for every partner

Cyber security

CEA Regulations apply from 1 April 2027

Privacy and consent

DPDP Rules apply from 13 May 2027

Regulatory reporting

No more re-keying for SERC filings

AI-ready operations

Clean, signed, machine-readable data

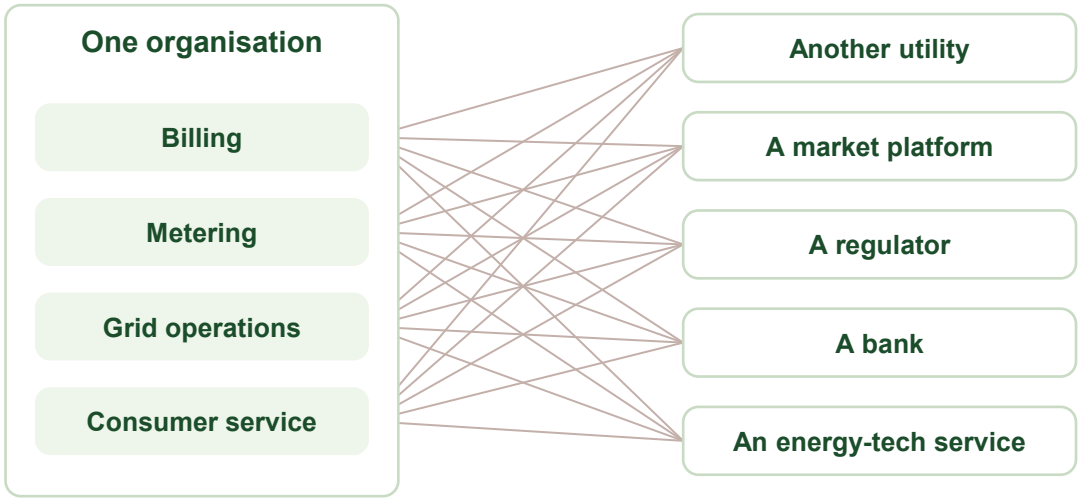
Consumer agency

Prove usage without paperwork

THE SOLUTION

One adapter at the edge of each system

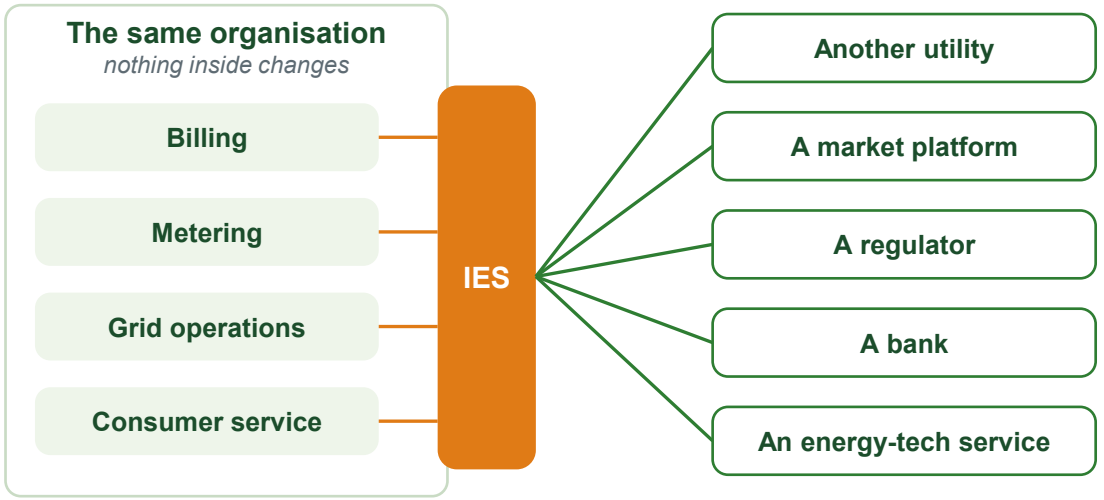
WITHOUT A COMMON EDGE



5 partners × 4 systems = 20 integrations

rebuilt at every refresh

WITH THE IES ADAPTER



Inside: your systems. Outside: IES.

5 partners = 1 adapter

Fitted once · Holds no data · Certified

DEFINITION

What IES is

> **Digital public infrastructure that lets energy systems exchange data, using common standards**

Common rules, not a platform

Any system can implement them

Data stays where it is

No central data store

Only the interface changes

Like UPI for payments

CLEARING UP CONFUSION

What IES is not

- Not a central database
- Not a replacement for DISCOM systems
- Not a new regulator
- Not a product to buy
- Not a grid control system
- Not a mandate to change internal systems

Three steps to share data

1

Register

Get a digital identity and join a shared directory

W3C Decentralised Identifiers

2

Discover

Find and verify the other party; agree terms

Beckn protocol

3

Exchange

Send data in agreed formats, with a receipt

DLMS · IEEE 2030.5 · OpenADR

STANDARDISED DATA

Common data standards

Taxonomy
Settles the words

Schemas
Settles the structure

API specifications
Settles the interface

Verifiable credentials
Settles the proof

Existing standards first, in this order:

BIS › CEA › IEC › IEEE › W3C · IETF · OCPP › New IES spec (last resort)

P2P ENERGY TRANSACTIONS

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P2P energy trading on IES

1

Discover and match

Buyer and seller find each other across apps

2

Contract

Signed and recorded on a shared ledger

3

Inject and consume

Checked against sanctioned load

4

Meter and signal

Smart-meter actuals signed by the DISCOM

5

Settle

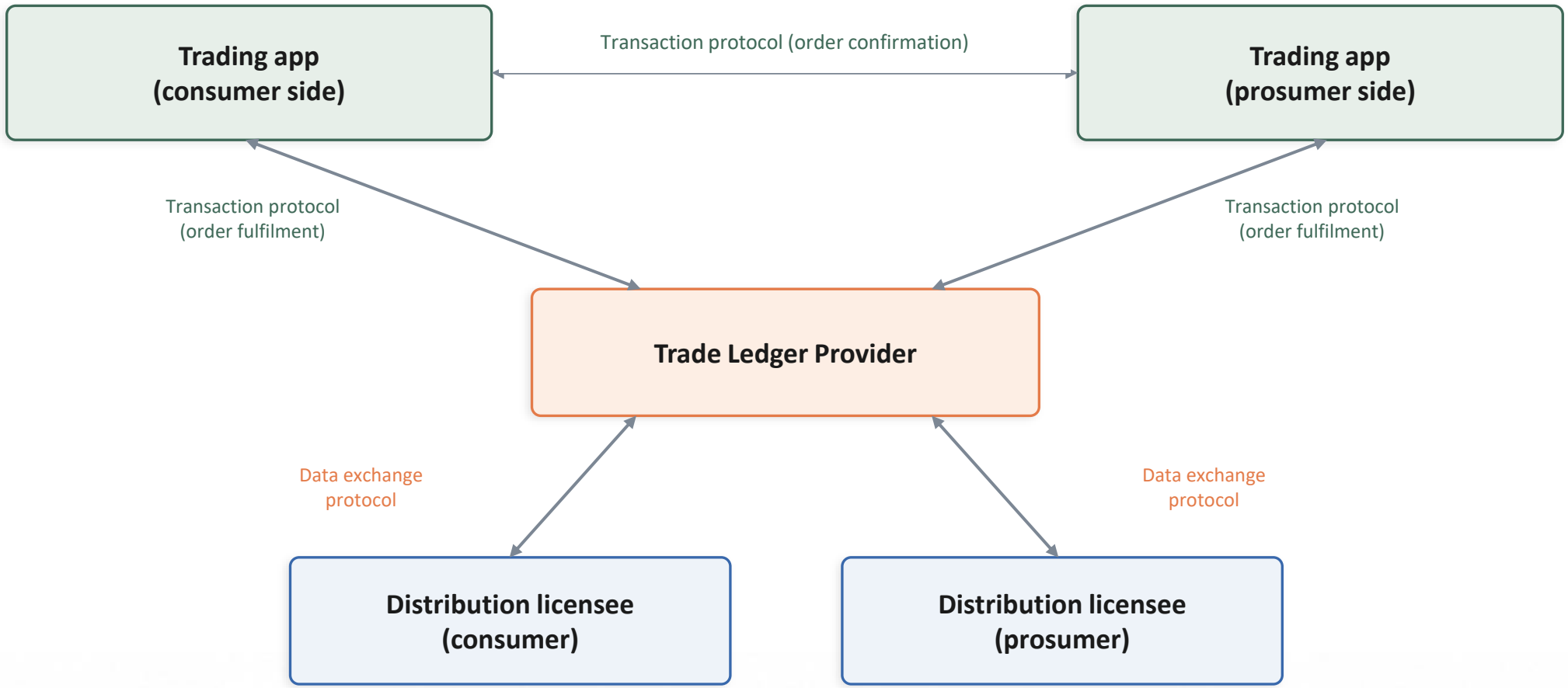
Paid against actuals; wheeling via billing

Live at BRPL, TPDDL and PVVNL

First inter-state trade, Feb 2026: a Meerut solar farmer sold power to a Delhi tailoring shop

P2P ENERGY TRANSACTIONS

How a P2P trade flows



P2P ENERGY TRANSACTIONS

Why this model can be replicated

1 **One shared ledger**

No bilateral integrations between platforms and DISCOMs

2 **Simple exchange layer for DISCOMs**

Allocation logic sits at the ledger

3 **One protocol for all use cases**

One ledger provider

4 **No central controller**

Signed receipts settle payments and disputes

OTHER USE CASES UNLOCKED

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Four use cases unlocked by standardised data exchange

01 IES schema layer

SCHEMA	
S1	ElectricityCredential
S2	MeterData
S3	MeterDataCredential
S4	MeterDataRequest
S5	MeterDataRequestCredential
S6	ArrFiling *in progress*
S7	OutageNotification *in progress*

02 Use cases enabled

USE CASE	
1	**Smart Meter Data Exchange** Published, consented meter-data exchange
2	**Energy Passport** Signed connection, meter and solar record
3	**Meter Digest** Signed consumption-history summary
4	**DER Visibility** PII-free grid-side view of DER

03 Schema dependency map

USE CASE	S1	S2	S3	S4	S5	S6	S7
Meter data exchange	●	✓	✓	✓	✓	●	●
Energy passport	✓	●	●	●	●	●	●
Meter digest	●	✓	✓	●	●	●	●
DER visibility	✓	●	●	●	●	●	●

USE CASES IN THE FIELD

Four DISCOMs live in 30 days

Tata Power

Mumbai

- Energy Passport
- Smart Meter Data Exchange
- DER Asset Characterisation

APEPDCL

Visakhapatnam

- Energy Passport
- My Meter Data
- Smart Meter Data Exchange

DGVCL

Surat

- Energy Passport
- Net-Meter Grid Impact

PVVNL

Meerut

- Energy Passport + Meter Digest
- DER Visibility
- Outage Publication (proposed)

~25 people · 3 days on site · 30-day build ·

NEW USE CASES

Anyone can propose a new schema

1

Open

'Contribute to IES' on the REC website

2

Submit

Each submission opens a public GitHub issue

3

Review

Chief architect team reviews it in the open

4

Publish

Versioned schema on IES GitHub and GitBook

One master vocabulary: no schema redefines a field

recindia.nic.in/india-energy-stack

NEW USE CASES

From pilot to all-India rollout

PILOT			VALIDATE	SCALE
1	2	3	4	5
Select pilot DISCOMs	**Onboard TSPs**	**Implement at pilots**	**Validate results**	**All-India rollout**
2–3 DISCOMs of different sizes	Schema, sandbox and guide	Built with real consumers	Checked against gate criteria	DISCOMs join in cohorts

ADOPTING IES

How a DISCOM adopts IES

1

Commit

Name an owner; pick a first use case

2

Assess

Check systems, data flows and contracts

3

Build on site

Team fits the adapter with the DISCOM

4

Go live

Certified in the sandbox, then live

Support from REC at every step

Forward deployment team

Specifications and guide

Sandbox and certification

Shared infrastructure

PMU and IES Cell

RECOMMENDATIONS

Next steps for DISCOMs

1 **Join the next cohort**

Name an owner; start with one use case

2 **Run IES and cyber readiness together**

3 **Issue Energy Passports**

A signed consumer record through DigiLocker

4 **Shape the standard**

Propose schemas

THANK YOU

For discussions/suggestions/queries email: ***dum@indiasmartgrid.org***

www.dumindia.in

India Energy Stack: ies@recindia.com · recindia.nic.in/india-energy-stack · ies.recindia.org.in/docs

